

ST 3011 Statistical Programming: Individual Contribution Report

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1. Project Overview

In the fishing industry normally before go to the sea the boat owner gives the fisherman a target to achieve during the trip, otherwise that trip will be non-profitable one, if it hasn't reached the targets as the total cost for the trip was not covered. Many experienced fishermen do this from their own heads but I have created an app so they can just put the values and get the results.

2. Modeling Components

- **Group project Part (Regression):** A Random Forest Regressor predicts total_cost_LKR based on engine HP, distance, crew size and duration, etc.
- **Individual Part (Classification & Simulation):**
 1. First, I created a pie chart which shows the sea condition. This was done as an innovative approach. I used Beaufort Wind Scale and calculated a new value using wave length and windspeed and then created a new variable called sea condition with 3 categories Calm, Moderate and Rough. From this the fisherman can see whether it is safe to go.
 2. The main individual contribution is, (Catch Target Simulator) I built a new feature that converts predicted costs into a Target Catch(kg). It tells the fisherman exactly how many Kilograms they must catch to pay for the trip. They just have to put the current market price of 1kg of the separate fish, and the target Kg of the fish will be calculated. If they pass the target that's where they start to gain the profit. Target covers the total cost, there on words they can earn profits.

3. Interactive Visualizations (New)

- **Live Safety Pie Chart:** The chart slice pops out when the user moves weather sliders.
- **Species Weight Histogram:** Added a selector to view historical catch ranges for Yellowfin, Skipjack, and Marlin.

4. Interpretation & Individual Value

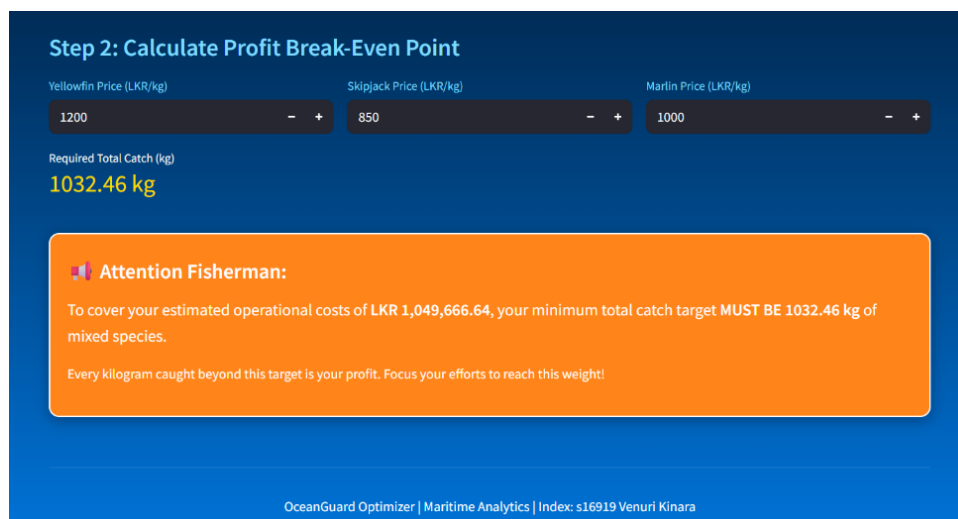
The app simplifies complex statistics for non-technical users. Instead of just showing a cost number, the simulator provides a direct goal: *"Your target is X kg."*

Limitation: The simulator assumes market prices remain stable throughout the trip, which may vary in reality.

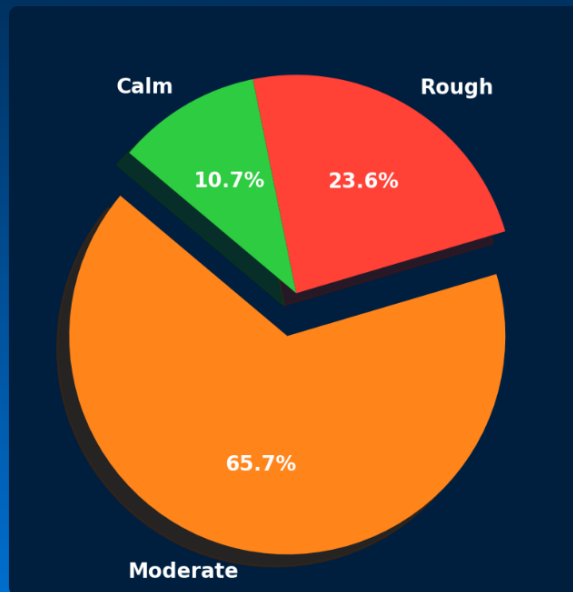
Group Project my work (Total cost prediction)



New work Individual project



Fleet Condition Distribution



Live Sea Safety Tester

Current Wind Speed (kph)



Current Wave Height (m)



STATUS: Moderate - PROCEED WITH CAUTION

Condition Reference Guide

- **Calm:** Wave < 1.0m & Wind < 15kph
- **Moderate:** Wave 1.0-2.0m or Wind 15-30kph
- **Rough:** Wave > 2.0m or Wind > 30kph

Species Weight Distribution (Historical)

Select Species to View Weight Range:

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